SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0701

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Debugging and Optimisation Task

Code of Conduct:

I ___Sk. Parvez Ahamed (192411084)__ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Sk. Parvez Ahamed*

1. IPv4 Subnet Configuration Error

Given configuration:

- IP address: 192.168.10.65
- Subnet mask: 255.255.255.192 (/26)
- Default gateway: 192.168.10.1
- Intended subnet: 192.168.10.0/26

Step 1: Identify the subnet

A /26 subnet contains 64 total IP addresses, with a block size of 64 in the last octet.

The subnet ranges within 192.168.10.0/24 are:

- 192.168.10.0 – 192.168.10.63
- 192.168.10.64 – 192.168.10.127
- 192.168.10.128 – 192.168.10.191
- 192.168.10.192 – 192.168.10.255

The configured IP address 192.168.10.65 belongs to 192.168.10.64/26, not 192.168.10.0/26.

Step 2: Network details

Property	**Intended subnet**	**Configured IP's subnet**
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Usable hosts	62	62

Step 3: Verify the IP address and gateway

- **IP address** 192.168.10.65**:** Valid host address in 192.168.10.64/26, but invalid for the intended subnet.
- **Gateway** 192.168.10.1**:** Valid host address in 192.168.10.0/26, but outside the workstation's configured subnet.

The workstation and gateway are in different /26 subnets, so the workstation cannot use that gateway as its normal local default gateway.

Step 4: Identify the configuration errors

1. The workstation has been assigned an IP address from the wrong subnet.
2. The gateway is not in the workstation's configured subnet.
3. The workstation's addressing does not match the intended network.

Step 5: Correct and optimize the configuration

Since the intended subnet is 192.168.10.0/26, correct the workstation settings as follows:

Setting	**Correct configuration**
IP address	192.168.10.2
Subnet mask	255.255.255.192
Default gateway	192.168.10.1
Network address	192.168.10.0
Broadcast address	192.168.10.63

The gateway 192.168.10.1 must be configured on the router interface connected to this subnet.

Step 6: Calculate usable host addresses

$$\text{Usable hosts} = 2^{32-26} - 2$$

$$= 2^6 - 2 = \boxed{62}$$

Answer: The corrected /26 subnet supports 62 usable host addresses.

2. IPv6 Addressing and Subnet Mismatch

Given addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Step 1: Expand both IPv6 addresses

IPv6 addresses contain eight groups of four hexadecimal digits.

Device A

Compressed:

2001:db8::ff00:42:8329/64

Expanded:

2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B

Compressed:

2001:db8:0:0:1::1/64

Expanded:

2001:0db8:0000:0000:0001:0000:0000:0001/64

Step 2: Determine the network prefix

A /64 prefix identifies the first four 16-bit groups.

Property	**Device A**	**Device B**
IPv6 address	2001:db8::ff00:42:8329	2001:db8:0:0:1::1

Prefix length	/64	/64
Network prefix	2001:db8:0:0::/64	2001:db8:0:0::/64
Interface identifier	0000:ff00:0042:8329	0001:0000:0000:0001

Step 3: Are both devices in the same subnet?

Yes. Both devices belong to the same /64 **subnet.**

The first four groups are identical in both addresses:

2001:0db8:0000:0000

Therefore, there is no IPv6 subnet-prefix mismatch in the given configuration.

Step 4: Identify possible causes of communication failure

Since the prefixes match, investigate other potential causes:

- Incorrect IPv6 default gateway or router advertisements.
- Duplicate IPv6 addresses.
- Firewall rules blocking ICMPv6 or application traffic.
- Incorrect VLAN assignment or Layer-2 connectivity.
- Neighbor Discovery Protocol (NDP) failure.
- Router or switch configuration errors.

Step 5: Optimize the IPv6 configuration

Both addresses can remain unchanged if they are unique and correctly configured.

For example:

Setting	**Device A**	**Device B**
IPv6 address	2001:db8::ff00:42:8329/64	2001:db8::1/64

Network prefix	2001:db8:0:0::/64	2001:db8:0:0::/64
Default gateway	Same valid subnet router	Same valid subnet router

The suggested Device B address is an alternative, not a necessary correction. Ensure that 2001:db8::1 is not already assigned to another device.

Step 6: Calculate the number of available addresses

A /64 leaves 64 bits for the interface identifier.

$$\text{Total addresses} = 2^{128-64} = 2^{64}$$

$$\boxed{18,446,744,073,709,551,616}$$

Answer: The subnet has 2^{64}possible interface identifiers. IPv6 does not subtract two addresses for network and broadcast addresses as IPv4 does.

Note: 2001:db8::/32 is reserved for documentation and examples, so these addresses should not be used as production public IPv6 addresses.

3. IPv4 Subnet Allocation and VLSM Optimization

Given network: 192.168.1.0/24

The organization uses /26, /27, and /28 subnets. Since the exact departmental requirements and existing subnet assignments have not been provided, the following is an example of a valid allocation that illustrates the calculations and optimization method.

Step 1: Calculate subnet capacities

Prefix	**Subnet mask**	**Total IP addresses**	**Usable host addresses**
/26	255.255.255.192	64	62

/27	255.255.255.224	32	30
/28	255.255.255.240	16	14

Formula:

$$\text{Usable hosts} = 2^{32-\text{prefix}} - 2$$

Step 2: Example of a valid subnet allocation

Allocate the largest subnet first, followed by smaller subnets.

Subnet	**Network address**	**Broadcast address**	**Valid host range**	**Usable hosts**
Department A /26	192.168.1.0	192.168.1.63	192.168.1.1 – 192.168.1.62	62
Department B /27	192.168.1.64	192.168.1.95	192.168.1.65 – 192.168.1.94	30
Department C /28	192.168.1.96	192.168.1.111	192.168.1.97 – 192.168.1.110	14

These three subnets do not overlap and are correctly aligned.

Step 3: Calculate unused addresses

The allocated address ranges are:

- /26: 64 addresses
- /27: 32 addresses
- /28: 16 addresses

Total allocated:

$$64 + 32 + 16 = 112$$

Total addresses in the /24:

$$2^{32-24} = 256$$

Unused addresses:

$$256 - 112 = \boxed{144}$$

The unused address range in this example is:

192.168.1.112 – 192.168.1.255

This range contains 144 total IP addresses.

Step 4: Identify inefficient subnet allocations

Subnet allocation is inefficient when:

- A department receives far more host addresses than it needs.
- A department receives fewer usable addresses than required.
- Subnets overlap or are not aligned to their proper boundaries.
- Large subnets are assigned to small departments while larger departments lack sufficient addresses.
- Network and broadcast addresses are mistakenly assigned to computers.

For example, a department requiring 12 hosts can use a /28, which provides 14 usable host addresses. Assigning it a /26 would provide 62 usable hosts and waste 50 host addresses, assuming no additional growth requirement.

Step 5: Recommend VLSM optimization

Variable Length Subnet Masking (VLSM) assigns different prefix lengths according to each department's host requirements.

Recommended process:

1. List the number of hosts required by each department, including expected growth.
2. Sort departments from largest to smallest.

3. Select the smallest subnet that accommodates each department.
4. Allocate subnets from the beginning of the address block, respecting subnet boundaries.
5. Verify that there are no overlaps and that enough space remains for future expansion.

For example:

Required hosts	Recommended prefix	Usable hosts
50	/26	62
25	/27	30
12	/28	14
6	/29	6

The exact optimized plan depends on the real number of devices in each department. A /29 provides six usable IPv4 host addresses, which may be insufficient if a router interface and other infrastructure devices must also be accommodated.

Conclusion: VLSM reduces address waste by assigning each department an appropriately sized subnet while preserving contiguous space for future use.

4. IPv6 Addressing and Subnet Mismatch

This question contains the same IPv6 addresses as Question 2, so the analysis is identical.

Step 1: Expanded addresses

- **Device A:** 2001:0db8:0000:0000:0000:ff00:0042:8329/64
- **Device B:** 2001:0db8:0000:0000:0001:0000:0000:0001/64

Step 2: Network prefixes

Both use the same first four groups:

2001:0db8:0000:0000

Therefore, both belong to:

2001:db8:0:0::/64

Step 3: Identify the mismatch

There is **no subnet-prefix mismatch**. Both devices are on the same /64 subnet.

The communication failure should instead be investigated through NDP, firewall rules, VLAN configuration, duplicate addresses, or other Layer-2 and Layer-3 connectivity problems.

Step 4: Optimized addressing

An example configuration is:

Device	IPv6 address	Prefix
A	2001:db8::ff00:42:8329	/64
B	2001:db8::1	/64

Both addresses are in the same subnet, assuming the addresses are unique.

Step 5: Available host addresses

$$2^{64} = \boxed{18,446,744,073,709,551,616}$$

This is the number of possible interface identifiers in a /64 subnet.

5. Router Routing Table Debugging

Given routing table:

Destination network	Next hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3

Default route (0.0.0.0/0)	10.0.0.1

Destination IP address: 192.168.2.100

Step 1: Identify the matching route using longest-prefix match

The destination IP 192.168.2.100 belongs to the network:

192.168.2.0/24

This network covers addresses from 192.168.2.0 through 192.168.2.255.

The router selects the route with the longest matching prefix. The /24 route is more specific than the default /0 route.

Step 2: Determine the next-hop router

The matching entry is:

Property	**Result**
Destination	192.168.2.100
Matching route	192.168.2.0/24
Selected next hop	10.0.0.3

Therefore, the router forwards the packet to 10.0.0.3, assuming the route is active and the next hop is reachable.

Step 3: Verify the destination subnet

The destination address is valid within 192.168.2.0/24.

- Network address: 192.168.2.0
- Broadcast address: 192.168.2.255
- Valid host range: 192.168.2.1 – 192.168.2.254
- Usable hosts: 254

Thus, the destination IP belongs to the intended subnet.

Step 4: Identify possible routing errors

The routing table contains a matching route, so the entries shown do not indicate an obvious destination-prefix error. Investigate the following possibilities:

- The next-hop address 10.0.0.3 is unreachable.
- The next-hop router lacks a route to the destination network.
- The return path is missing or incorrect.
- An access-control list or firewall blocks the traffic.
- The destination host is offline or has an incorrect gateway.
- The route is configured but inactive in the actual routing table.

The next-hop addresses must also be reachable through an appropriate directly connected or recursively resolved route.

Step 5: Optimize the routing table

The existing table is already concise and includes a specific route for each destination network. No change is required solely on the basis of the supplied entries.

Recommended improvements:

1. Verify that 10.0.0.2, 10.0.0.3, and 10.0.0.1 are reachable.
2. Confirm that the next-hop routers have valid return routes.
3. Remove duplicate or obsolete routes if they exist in the full routing table.
4. Use dynamic routing protocols such as OSPF when appropriate for a larger network.
5. Maintain a valid default route for destinations not covered by more specific entries.

Step 6: Identify the default route

If no more specific destination route matches, the router uses:

0.0.0.0/0 → 10.0.0.1

The default route acts as the fallback route for IPv4 destinations that do not match any more specific route.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation